

Boundary Phase Compression in a Causal Zero-Repair Transport Recursion

An exact six-dimensional quotient behind the $r = 0$ obstruction

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Abstract

We isolate a finite-dimensional latent degree of freedom in a causal zero-repair transport recursion with coarse block laws

$$P_b = \frac{1}{64}(1, 12, 38, 12, 1), \quad Q_b = \frac{1}{64}(0, 16, 32, 16, 0).$$

Their primitive defect is exactly

$$P_b(t) - Q_b(t) = \frac{(1-t)^4}{64}.$$

At the terminal boundary $r = 0$, the natural interior continuation creates precisely two illegal transports, with masses $3/128$ and $1/128$, hence total debt $1/32$.

The main result of this note is an exact support-independent compression theorem. For the nine natural boundary observables consisting of the first and third critical child Hall prefixes on branches $x \in \{0, 2, 4\}$, together with the two donor capacities and their shared overlap, the image of the zero-marginal transport kernel has dimension exactly six for every finite parent support $m \geq 2$. Three of the nine raw observables are already determined by the parent marginals. A six-dimensional lower bound is proved by six explicit elementary transport circulations whose readout matrix has determinant $2^{20}/19^2 \neq 0$.

The computational relation compiler that led to this reduction is used only as a discovery and regression instrument. An exact rational scan over supports $2, \dots, 12$ reproduces rank six throughout; a finite relation atlas over 256 concrete states yields 4590 feasible and 18 infeasible one-step support queries, with all 18 infeasibilities coming from the single degenerate state $B(t) = 1$. No unknown solver statuses occur.

The result does not prove recursive boundary closure or the desired logarithmic asymptotic bound. It replaces the full witness-tree obstruction by a concrete six-dimensional boundary-phase compatibility problem.

1 Local transport model

Let

$$P_b(x) = \frac{1}{64}(1, 12, 38, 12, 1)_x, \quad x = 0, \dots, 4,$$

and

$$Q_b(y) = \frac{1}{64}(0, 16, 32, 16, 0)_y, \quad y = 0, \dots, 4.$$

Equivalently,

$$P_b(t) = \frac{1 + 12t + 38t^2 + 12t^3 + t^4}{64}, \quad Q_b(t) = \frac{16t + 32t^2 + 16t^3}{64}.$$

The elementary identity

$$P_b(t) - Q_b(t) = \frac{(1-t)^4}{64} \quad (1)$$

will be referred to as the *primitive defect*.

Let

$$B(t) = \sum_{j=0}^m b_j t^j, \quad b_j \geq 0, \quad \sum_j b_j = 1,$$

be a finite buffer law. A one-step causal zero-repair witness is a nonnegative transport

$$\pi(x, u) \geq 0$$

with row marginals

$$\sum_u \pi(x, u) = P_b(x) \quad (2)$$

and column marginals

$$\sum_x \pi(x, u) = (Q_b * B)(u), \quad (3)$$

subject to the causal support constraint

$$\pi(x, u) = 0 \quad \text{for } u < x. \quad (4)$$

Whenever $P_b(x) > 0$, the residual child law on branch x is read directly from the transport:

$$B_x(j) = \frac{\pi(x, x+j)}{P_b(x)}. \quad (5)$$

Thus the child laws are not independent variables. They are boundary readouts of one local transport object.

For a fixed parent support m , let M_m denote the linear map that sends a transport array to all row and column marginals. Its kernel

$$\mathcal{K}_m := \ker M_m$$

is the space of zero-marginal circulations: internal transport changes invisible to the parent and child marginal totals.

2 The terminal boundary obstruction

The interior terminal splitting rule is exact for $r \geq 1$, but at $r = 0$ two residual -1 channels become illegal. They occur at

$$(x, u) = (2, 1), \quad (x, u) = (4, 3).$$

Including the terminal weight $1/2$, their masses are

$$\frac{1}{2} \cdot \frac{38}{64} \cdot \frac{3}{38} = \frac{3}{128}, \quad \frac{1}{2} \cdot \frac{1}{64} = \frac{1}{128}.$$

Hence the total local repair debt is

$$\frac{3}{128} + \frac{1}{128} = \frac{1}{32}. \quad (6)$$

The point of the present note is not to construct the full recursive repair. It is to identify exactly how much latent transport information the boundary can see.

3 Natural boundary readout

For any buffer law C , define the two critical prefix functionals

$$H_1(C) := \Pr_{Q_b * C}[U \leq 1] = \frac{1}{4} C_0, \quad (7)$$

$$H_3(C) := \Pr_{Q_b * C}[U \leq 3] = C_0 + \frac{3}{4} C_1 + \frac{1}{4} C_2. \quad (8)$$

For a feasible transport π , define the donor capacities

$$D_{21}(\pi) := \sum_{\substack{x \leq 1 \\ u \geq 2}} \pi(x, u), \quad (9)$$

$$D_{43}(\pi) := \sum_{\substack{x \leq 3 \\ u \geq 4}} \pi(x, u), \quad (10)$$

$$D_{\text{sh}}(\pi) := \sum_{\substack{x \leq 1 \\ u \geq 4}} \pi(x, u). \quad (11)$$

The first two are the total donor pools naturally available to repair the illegal cells $(2, 1)$ and $(4, 3)$. The third is their shared donor sector.

The raw boundary readout is the linear map

$$\mathcal{R}_m(\pi) := (H_1(B_0), H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{21}, D_{43}, D_{\text{sh}}). \quad (12)$$

Only the restriction of $\mathcal{R}_m$ to $\mathcal{K}_m$ represents genuine latent phase: any direction fixed by row and column marginals is not hidden transport freedom.

4 Exact phase-compression theorem

Theorem 1 (Support-independent boundary phase). *For every finite parent support $m \geq 2$,*

$$\boxed{\dim \mathcal{R}_m(\mathcal{K}_m) = 6.}$$

Equivalently, among the nine raw boundary observables in (12), precisely six independent linear degrees of freedom remain after quotienting by the row and column marginals.

Proof. We prove the upper and lower bounds separately.

Upper bound. Three raw outputs are already fixed by the parent marginals.

First, the column $u = 0$ has zero mass because $Q_b(0) = 0$. Hence

$$\pi(0, 0) = 0,$$

and therefore

$$H_1(B_0) = 0. \quad (13)$$

Second, since the total row mass for $x \leq 1$ is

$$P_b(0) + P_b(1) = \frac{13}{64},$$

while every unit of column mass in $u \leq 1$ necessarily comes from rows $x \leq 1$, we obtain

$$\begin{aligned} D_{21} &= \frac{13}{64} - \Pr_{Q_b * B}[U \leq 1] \\ &= \frac{13}{64} - \frac{1}{4}B_0. \end{aligned} \tag{14}$$

Third, the total row mass for $x \leq 3$ is

$$\sum_{x=0}^3 P_b(x) = \frac{63}{64},$$

and every column $u \leq 3$ is inaccessible to row $x = 4$. Thus

$$\begin{aligned} D_{43} &= \frac{63}{64} - \Pr_{Q_b * B}[U \leq 3] \\ &= \frac{63}{64} - \left(B_0 + \frac{3}{4}B_1 + \frac{1}{4}B_2 \right). \end{aligned} \tag{15}$$

Hence $H_1(B_0)$, D_{21} , and D_{43} have no independent variation on the zero-marginal kernel. Therefore

$$\dim \mathcal{R}_m(\mathcal{K}_m) \leq 6.$$

Lower bound. Let $E_{x,u}$ denote the unit transport cell. Whenever all four cells are causally admissible, define the elementary plaquette circulation

$$C_{x,u} := E_{x,u} - E_{x,u+1} - E_{x+1,u} + E_{x+1,u+1}. \tag{16}$$

Each $C_{x,u}$ has zero row sums and zero column sums, hence belongs to $\mathcal{K}_m$.

For every $m \geq 2$, the six plaquettes

$$C_{0,1}, \quad C_{1,2}, \quad C_{1,3}, \quad C_{1,4}, \quad C_{3,4}, \quad C_{3,5}$$

are all available, because their largest column index is $6 \leq m + 4$.

Evaluate them on the six candidate phase coordinates

$$\Phi := (H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{\text{sh}}). \tag{17}$$

With the plaquettes in the order above, the resulting 6×6 matrix is

$$A = \begin{pmatrix} 32 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8}{19} & 0 & 0 & 0 & 0 \\ 0 & -\frac{8}{19} & -\frac{16}{19} & -\frac{8}{19} & 0 & 0 \\ 0 & 0 & 0 & 0 & -16 & 0 \\ 0 & 0 & 0 & 0 & -16 & -32 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}. \tag{18}$$

A direct calculation gives

$$\det A = \frac{2^{20}}{19^2} \neq 0. \tag{19}$$

Therefore the six coordinates in (17) are linearly independent on $\mathcal{K}_m$, so

$$\dim \mathcal{R}_m(\mathcal{K}_m) \geq 6.$$

Combining both inequalities proves the theorem. $\square$

Corollary 1 (Individual donor budgets are not phase). *The total donor capacities D_{21} and D_{43} are completely determined by the parent low-edge profile:*

$$D_{21} = \frac{13}{64} - \frac{1}{4}B_0,$$

$$D_{43} = \frac{63}{64} - \left(B_0 + \frac{3}{4}B_1 + \frac{1}{4}B_2\right).$$

The donor-related latent degree of freedom visible to the boundary is therefore not the individual pool size but the shared allocation coordinate D_{sh} .

5 Interpretation

The theorem changes the natural formulation of the boundary problem.

A naive picture asks whether enough donor mass exists to pay the local debt $1/32$. But corollary 1 shows that the two total donor budgets are already fixed by the parent marginals. They are not hidden variables.

The genuine latent information is six-dimensional. One convenient basis is

$$\Phi = (H_3(B_0), H_1(B_2), H_3(B_2), H_1(B_4), H_3(B_4), D_{\text{sh}}).$$

Five coordinates measure how a local transport choice changes the future low-edge/Hall geometry of the critical children. The sixth measures how much donor capacity lies in the sector shared by the two repair channels.

Thus the remaining obstruction is better stated as follows:

Can one choose the six-dimensional transport phase at every boundary encounter so that the local debt is repaired and the resulting critical descendants remain recursively compatible with the same phase calculus?

This is substantially smaller than the original witness-tree formulation. It is still nontrivial: local feasibility alone does not imply recursive compatibility.

6 Computational discovery and audit

The theorem above is exact and independent of the numerical relation atlas. The computation served two roles: discovery of the correct quotient and regression checking of the algebra.

The compiler was run with parent support 12, three requested closure rounds, a cap of 256 concrete states, and eight allowed refinement rounds. It performs no tree search and uses no random objectives. Its local object is the exact nonnegative transport polytope defined by (2)–(4).

The 18 infeasible support queries should not be interpreted as 18 distinct boundary obstructions. All 18 arise from the same concrete state

$$B(t) = 1,$$

for which the one-step transport is exactly impossible: row $x = 4$ requires mass $P_b(4) = 1/64$ at columns $u \geq 4$, whereas $Q_b * 1 = Q_b$ has no mass at $u \geq 4$. Since each concrete state is queried in 18 min/max support directions, this single infeasible parent accounts for all 18 failures.

Quantity	Result
Primitive defect identity	exact
Boundary debt	1/32
Raw boundary outputs	9
Exact phase rank at support 12	6
Exact phase-rank scan, supports $2, \dots, 12$	6 throughout
Concrete states retained	256
One-step support queries	4608
FEASIBLE	4590
INFEASIBLE	18
UNKNOWN	0
Parent-side CEGAR refinements	0
Exposed atlas transitions	8926

Table 1: Computational audit of the boundary semantic compiler.

The state cap was reached. Consequently, the 256 concrete states and 8926 exposed transitions are a finite exploratory atlas, not a closed automaton. They are not used in the proof of theorem 1.

The absence of parent-side CEGAR refinements also has a precise interpretation. The initial parent interface already contains

$$B_0, \quad B_0 + \frac{3}{4}B_1 + \frac{1}{4}B_2, \quad \mathbf{E}[J].$$

The first two determine the phase-invariant donor capacities in (14)–(15). Hence no additional parent functional is forced by the phase-invariant part of the chosen boundary readout. The unresolved information lives in the internal six-dimensional transport phase, not in an omitted scalar parent statistic.

7 Relation to the logarithmic upper-bound program

The dyadic recursion

$$f_0(t) = t, \quad f_{k+1}(t) = \frac{1}{2}t + \frac{1}{2}f_k(t)^2$$

has the elementary derivative identity

$$f'_{k+1}(1) = f'_k(1) + \frac{1}{2},$$

and therefore

$$f'_k(1) = 1 + \frac{k}{2}.$$

After the parity substitution $t = z^2$, the associated mean grows only linearly in k , while the natural scale grows dyadically.

The missing step in turning this mechanism into a logarithmic upper bound is not the mean calculation. It is recursive causal closure at the boundary $r = 0$. The present theorem does not provide that closure. It shows, however, that the boundary-visible part of the local transport ambiguity is finite-dimensional and support-independent.

This suggests that the relevant object is not an unbounded witness tree but a positive composition law on a six-dimensional latent phase.

8 Next exact target

A natural next step is to characterize the feasible phase set

$$\mathcal{P}(B) := \{\Phi(\pi) : \pi \in \mathcal{T}(B)\} \subset \mathbb{R}^6,$$

where $\mathcal{T}(B)$ denotes the exact one-step transport polytope and Φ is the basis in (17).

The desired recursive statement would not require a unique phase. It would require a positive, history-compatible selection rule.

Conjecture 1 (Boundary phase closure). *There exists a scale-stable admissible family of phase sets, or an equivalent finite relation calculus on Φ , such that every boundary state generated by the dyadic recursion admits a feasible phase whose critical descendants remain in the same admissible family.*

A proof of this statement, in a form strong enough to compose through the causal recursion, would close the remaining local boundary obstruction in the logarithmic upper-bound strategy.

9 Scope

The exact theorem in this note concerns the stated nine-dimensional local boundary readout and its quotient by the transport marginal kernel. It does not assert that every aspect of the full causal witness admits a six-dimensional representation.

Likewise, the finite computational atlas is not a proof of recursive closure. Floating-point LP support values and rational reconstructions are discovery data unless separately certified. The exact rank theorem is independent of those numerical values.

The main conclusion is therefore deliberately narrow:

**The natural local $r = 0$ boundary readout has an exact,
support-independent six-dimensional latent phase.**

This turns a diffuse witness-tree obstruction into a concrete finite-dimensional composition problem.